CASE STUDY REPORT

Exploratory Data Analysis and Visualization of a Real-World Dataset

E-Commerce Customer Behavior Analysis and Churn Prediction Using Deep Learning

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ABSTRACT

E-commerce platforms collect customer data to understand purchasing behavior, improve customer satisfaction, and maintain long-term customer relationships. This project presents an exploratory data analysis and deep learning approach for analyzing e-commerce customer behavior and predicting customer churn. The dataset contains 5,630 customer records and 20 columns, including customer tenure, preferred login device, payment method, satisfaction score, order count, cashback amount, and churn status. Data preprocessing includes missing-value handling, duplicate checking, outlier treatment, categorical encoding, feature scaling, and dataset splitting. Exploratory data analysis is performed using visualizations to identify customer behavior patterns and possible churn indicators. A Multilayer Perceptron (MLP) neural network is trained using TensorFlow and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The model achieved a test accuracy of 91.39% and a ROC-AUC of 0.9669, demonstrating promising predictive performance on the test dataset.

Keywords: E-commerce, customer behavior, exploratory data analysis, customer churn, deep learning, neural networks, data visualization.

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1. INTRODUCTION

The growth of e-commerce has created opportunities for businesses to reach customers through online platforms. At the same time, online businesses face challenges in understanding customer preferences, maintaining satisfaction, and retaining customers over time.

Customer churn refers to a situation in which a customer stops engaging with a business or is classified as having left according to the dataset's churn definition. Identifying customers who may churn can help businesses investigate possible problems and develop suitable customer-retention strategies.

Exploratory Data Analysis (EDA) is an important step in data science. It involves examining a dataset using statistical summaries and visualizations to understand its structure, detect missing values, identify unusual observations, and discover relationships between variables.

Deep learning techniques can use multiple customer characteristics together to identify patterns associated with churn. In this project, an MLP neural network is developed to predict whether a customer belongs to the churn or non-churn class.

The project combines EDA, data preprocessing, visualization, and deep learning to demonstrate how customer data can support data-driven decision-making in e-commerce.

1.1 Problem Statement

E-commerce businesses need to understand customer behavior and identify customers who may stop using their services. Examining individual customer attributes alone may not reveal all relevant patterns.

Therefore, this project analyzes an e-commerce customer dataset and develops a deep learning classification model to predict customer churn based on available customer characteristics.

1.2 Objectives

The main objectives of this project are:

1. To load and understand a real-world e-commerce customer dataset.
2. To identify missing values, duplicate records, inconsistent data, and potential outliers.
3. To perform exploratory data analysis using statistical summaries and visualizations.
4. To study the relationship between customer characteristics and churn.
5. To encode categorical variables and standardize numerical features.
6. To divide the dataset into training, validation, and testing sets.
7. To develop an MLP deep learning model for customer churn prediction.
8. To perform basic hyperparameter tuning using validation performance.
9. To evaluate the model using suitable classification metrics and visualizations.
10. To discuss the usefulness and limitations of the results for customer-retention analysis.

1.3 Scope of the Project

The project focuses on customer-level churn classification using the available dataset. It examines factors such as tenure, preferred login device, payment method, order category, satisfaction score, complaints, and purchasing behavior.

The model predicts a churn class; it does not directly determine why an individual customer leaves. The findings can help identify patterns that deserve further investigation by an e-commerce business.

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2. DATASET DESCRIPTION

2.1 Dataset Overview

The dataset used in this project is an e-commerce customer churn dataset obtained from Kaggle.

Dataset details:

- Dataset file: E Commerce Dataset.xlsx

- Worksheet used: E Comm
- Number of records: 5,630
- Number of columns: 20
- Target variable: Churn
- Project type: Binary classification
- Implementation environment: Google Colab
- Programming language: Python

The Excel workbook also contains a Data Dict worksheet that describes the dataset fields. The E Comm worksheet is used for the analysis.

2.2 Description of Features

S. No.	Feature	Description
1	CustomerID	Unique customer identifier
2	Churn	Target variable indicating churn status
3	Tenure	Customer tenure
4	PreferredLoginDevice	Customer's preferred login device
5	CityTier	Customer's city classification
6	WarehouseToHome	Distance from warehouse to home
7	PreferredPaymentMode	Preferred payment method
8	Gender	Customer gender category
9	HourSpendOnApp	Time spent on the application
10	NumberOfDeviceRegistered	Number of registered devices
11	PreferedOrderCat	Preferred order category
12	SatisfactionScore	Customer satisfaction score
13	MaritalStatus	Customer's marital-status category
14	NumberOfAddress	Number of addresses registered
15	Complain	Indicator of whether the customer complained
16	OrderAmountHikeFromlastYear	Increase in order amount compared with the previous year
17	CouponUsed	Number of coupons used

18	OrderCount	Number of orders
19	DaySinceLastOrder	Days since the last order
20	CashbackAmount	Cashback amount associated with the customer

2.3 Target Variable

The target variable is Churn.

- **0 – Non-churn:** The customer belongs to the non-churn class.
- **1 – Churn:** The customer belongs to the churn class.

The project uses the target labels supplied in the dataset. These labels are not independently verified against future customer activity.

2.4 Tools and Technologies Used

Tool or Library	Purpose
Python	Main programming language
Google Colab	Development and execution environment
Pandas	Data loading, cleaning, and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical charts
Scikit-learn	Preprocessing, splitting, and evaluation
TensorFlow/Keras	Deep learning model development
Excel	Original dataset format
GitHub	Project version control and submission

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3. DATA EXPLORATION AND PREPROCESSING

Data preprocessing is necessary to ensure that the dataset is suitable for analysis and model training. The following steps were performed in the project.

3.1 Data Loading

The dataset was loaded from the E Commerce Dataset.xlsx file using Pandas. The E Comm worksheet was selected for analysis.

The dataset shape, column names, data types, first few rows, and statistical summaries were examined.

The initial dataset contained 5,630 records and 20 columns.

3.2 Handling Missing Values

Missing values were checked using the isnull() function. Seven features contained missing values.

Feature	Missing Values	Approximate Percentage
DaySinceLastOrder	307	5.45%
OrderAmountHikeFromlastYear	265	4.71%
Tenure	264	4.69%
OrderCount	258	4.58%
CouponUsed	256	4.55%
HourSpendOnApp	255	4.53%
WarehouseToHome	251	4.46%

Numerical missing values were handled using median imputation. Categorical missing values were handled using the most frequent category.

The median is less sensitive to extreme numerical observations than the mean. The most frequent category provides a simple replacement for missing categorical entries.

During model preparation, the imputation rules were fitted on the training data and then applied to the validation and test data to reduce data leakage.

3.3 Duplicate Record Checking

Duplicate records were checked using Pandas' duplicated() function.

The initial dataset contained **zero duplicate rows**. Therefore, no records were removed because of exact row duplication.

3.4 Column and Data Cleaning

Column names were stripped of unnecessary leading or trailing spaces. Accidental unnamed index columns were removed if present.

Leading and trailing spaces in text values were also removed. Numerical and categorical columns were identified using their data types.

The CustomerID column was excluded from the model inputs because it is an identifier rather than a meaningful behavioral measurement.

3.5 Outlier Detection and Treatment

Boxplots were used to identify possible outliers in numerical features.

The Interquartile Range (IQR) method was used to determine lower and upper limits:

[
IQR = Q_3-Q_1
]

[
Lower\ Bound = Q_1-1.5(IQR)
]

[
Upper\ Bound = Q_3+1.5(IQR)
]

Observations outside these limits are treated as potential outliers, not automatically as errors.

For model preprocessing, numerical values were capped at IQR limits learned from the training set. This reduces the influence of extreme values while retaining the associated customer records.

3.6 Encoding Categorical Variables

The dataset contains categorical features such as preferred login device, payment mode, gender, order category, and marital status.

These features cannot be directly processed as ordinary text inputs by the neural network. One-Hot Encoding was used to convert categorical values into numerical indicator columns.

The encoder was configured to ignore previously unseen categories during transformation.

Some categories in the original data have similar meanings but different labels, such as Phone and Mobile Phone, CC and Credit Card, or COD and Cash on Delivery. These labels could be standardized further in a future version of the project.

3.7 Feature Scaling

Numerical features were standardized using StandardScaler.

The transformation is:

[
z=\frac{x-\mu}{\sigma}
]

where:

- (x) is the original value.
- (\mu) is the mean calculated from the training data.
- (\sigma) is the standard deviation calculated from the training data.
- (z) is the standardized value.

Standardization places numerical features on comparable scales and can help neural-network training.

3.8 Training, Validation, and Testing Split

The dataset was split into three subsets:

Dataset Subset	Percentage of Total Data	Purpose

Training set	64%	Learn model parameters
Validation set	16%	Compare configurations and monitor performance
Test set	20%	Evaluate final model performance

Stratified splitting was used to preserve the churn-class proportions as closely as possible in each subset.

The preprocessing pipeline was fitted on training data only. The same fitted pipeline was then applied to the validation and test data.

3.9 Class Imbalance

The original target distribution was:

Churn Class	Number of Customers	Percentage
Non-churn (0)	4,682	83.16%
Churn (1)	948	16.84%
Total	**5,630**	**100%**

The dataset contains substantially more non-churn customers than churn customers.

To help address this imbalance during training, class weights were calculated from the training labels and supplied to the neural network.

Accuracy was not used as the only evaluation measure because a model can obtain high accuracy while performing poorly on the minority churn class.

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4. EXPLORATORY DATA ANALYSIS AND VISUALIZATION

Exploratory Data Analysis was performed to understand the structure of the dataset, examine the distribution of customer attributes, and investigate relationships with churn.

At least five EDA visualizations were created. The following sections explain the purpose and interpretation of the charts.

4.1 Visualization 1: Customer Churn Distribution

Chart type: Count plot

Purpose: To compare the number of customers in the churn and non-churn classes.

Observation: The dataset contains 4,682 non-churn customers and 948 churn customers. Approximately 83.16% belong to the non-churn class, while 16.84% belong to the churn class.

Insight: The target classes are imbalanced. This supports the use of class weights and evaluation metrics such as recall, precision, F1-score, and ROC-AUC in addition to accuracy.

4.2 Visualization 2: Distribution of Numerical Features

Chart type: Histograms

Purpose: To examine the distributions of numerical customer features, including tenure, distance from warehouse to home, app usage, order count, coupons used, and cashback amount.

Observation: The dataset contains numerical variables with different ranges and distributions. For example, tenure ranges from 0 to 61 in the supplied data, while cashback amount ranges from 0 to approximately 325.

Insight: The variables have different scales, making numerical scaling appropriate for the neural network. Histograms also help identify skewed distributions and values that may require further investigation.

4.3 Visualization 3: Categorical Feature Distribution

Chart type: Count plots

Purpose: To understand the distribution of categories in features such as preferred login device, payment mode, gender, preferred order category, and marital status.

Observation: Mobile Phone is the most frequently recorded preferred login device, with 2,765 records. Debit Card is the most frequent recorded payment label, with 2,314 records. Married customers form the largest marital-status group, with 2,986 records.

Insight: Customer behavior is represented by a combination of device preferences, payment choices, order categories, and demographic categories. These variables can be encoded and used as model inputs.

4.4 Visualization 4: Correlation Heatmap

Chart type: Correlation heatmap

Purpose: To examine pairwise linear correlations between numerical variables.

Observation: The heatmap displays positive, negative, and weak relationships between numerical features. The target variable can also be included to explore its numerical associations with the input features.

Insight: Correlation analysis helps identify potentially related variables and provides a preliminary understanding of the dataset. However, correlation alone does not establish causation or capture all nonlinear relationships.

Note: Specific correlation values should be reported from the generated heatmap rather than estimated visually or invented.

4.5 Visualization 5: Numerical Features Compared with Churn

Chart type: Boxplots grouped by churn status

Purpose: To compare the distributions of numerical customer characteristics between churned and non-churned customers.

Observation: In the supplied data, the mean tenure is approximately 11.50 for non-churn customers and 3.38 for churn customers. The mean order count is approximately 3.05 for non-churn customers and 2.82 for churn customers.

Insight: The tenure difference suggests that shorter-tenure customers may warrant closer investigation. The difference in average order count is smaller, showing why multiple variables should be considered together rather than relying on a single feature.

These are descriptive differences and do not independently prove that tenure or order count causes churn.

4.6 Visualization 6: Outlier Analysis

Chart type: Boxplots for numerical features

Purpose: To identify possible extreme observations in numerical variables.

Observation: The charts show the spread, median, interquartile range, and potential extreme observations for each numerical feature.

Insight: Outlier detection helps identify values that may influence model training. Extreme observations should be checked against the meaning of each variable before being treated as errors. The project uses IQR-based capping during model preprocessing.

4.7 Visualization 7: Categorical Features Compared with Churn

Chart type: Churn-rate comparisons across categories

Purpose: To examine whether churn rates differ across customer groups.

Observation: In the original data, the churn rate is approximately 31.7% among customers recorded as having complained, compared with 10.9% among customers without a recorded complaint. The churn rate is also approximately 26.7% among single customers and 11.5% among married customers.

Insight: Complaint status and marital-status category show differences in observed churn rates. These relationships may be useful for further investigation, but they do not establish causation. The results should be interpreted alongside sample sizes, other features, and potential confounding factors.

4.8 Overall EDA Findings

The exploratory analysis revealed several important characteristics of the dataset:

1. The dataset contains 5,630 records and 20 columns.
2. Seven numerical features contain missing values, with missing percentages ranging from approximately 4.46% to 5.45%.
3. No exact duplicate rows were detected.
4. The target variable is imbalanced, with churned customers representing approximately 16.84% of the dataset.
5. Customer tenure, complaint status, and purchasing behavior show descriptive differences between churn groups.
6. Categorical features contain labels that could be standardized further.
7. Data preprocessing is required before training the neural network.

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5. DEEP LEARNING MODEL DEVELOPMENT

5.1 Model Selection

A Multilayer Perceptron (MLP) was selected for customer churn prediction.

An MLP is a feed-forward artificial neural network consisting of an input layer, one or more hidden layers, and an output layer. It can learn nonlinear relationships among numerical and encoded categorical features.

The dataset is tabular rather than image- or sequence-based, so an MLP is a suitable starting point for this binary classification task.

5.2 Model Architecture

The model contains the following layers:

Layer	Configuration	Purpose
Input layer	Number of transformed input features	Receives preprocessed customer data
Hidden layer 1	Dense, 64 units, ReLU activation	Learns patterns from the inputs
Dropout layer 1	Dropout rate 0.20	Helps reduce overfitting
Hidden layer 2	Dense, 32 units, ReLU activation	Learns additional feature combinations
Dropout layer 2	Dropout rate 0.10	Adds regularization
Output layer	Dense, 1 unit, sigmoid activation	Produces churn probability

The model uses the Adam optimizer and binary cross-entropy loss.

5.3 Activation Functions

The ReLU activation function is used in the hidden layers:

$$
ReLU(x)=\max(0,x)
$$

ReLU introduces nonlinearity, allowing the network to learn more complex patterns.

The sigmoid activation function is used in the output layer:

$$
\sigma(x)=\frac{1}{1+e^{-x}}
$$

The sigmoid output represents a value between 0 and 1, interpreted as the model's estimated probability of churn for the positive class.

A threshold of 0.5 is used to convert probabilities into predicted classes in the reported evaluation.

5.4 Loss Function and Optimizer

Binary cross-entropy is used because the target contains two classes.

The Adam optimizer updates the model's weights to reduce the loss during training. The learning rate controls the size of these updates.

5.5 Model Training

The model was trained using the training set and monitored using the validation set.

The training configuration used:

- Maximum epochs: 50 for the initial model.
- Batch size: 32.
- Optimizer: Adam.
- Loss function: Binary cross-entropy.
- Evaluation metrics during training: Accuracy and AUC.
- Early stopping: Monitors validation loss with a patience of eight epochs.
- Class weighting: Used to account for the unequal target distribution.

Early stopping restores the weights associated with the best observed validation loss, reducing the risk of unnecessary training and overfitting.

5.6 Hyperparameter Tuning

Basic hyperparameter tuning was performed by comparing three configurations:

Experiment	Hidden Units	Learning Rate
1	32	0.001
2	64	0.001
3	64	0.0005

Each configuration was evaluated using validation loss, validation accuracy, and validation AUC.

The configuration with the highest validation AUC was selected for final testing. The test dataset was kept separate from the hyperparameter selection process.

The exact winning configuration should be recorded from the output of the Colab tuning cell.

5.7 Feature Analysis

The project examines the magnitude of the first dense layer's weights to produce an exploratory ranking of transformed input features.

This approach provides an initial view of how strongly the input features are weighted by the first layer. However, neural-network weights alone are not a definitive measure of feature importance because the model contains multiple layers and nonlinear transformations.

Permutation importance or SHAP-based analysis could provide a more interpretable investigation in future work.

6. MODEL EVALUATION METRICS

The final model was evaluated using the test dataset, which was not used to select the hyperparameters.

The following metrics were used.

6.1 Accuracy

Accuracy measures the proportion of correct predictions.

$$
Accuracy=\frac{TP+TN}{TP+TN+FP+FN}
$$

6.2 Precision

Precision measures the proportion of predicted churn cases that were actually churn cases.

$$
Precision=\frac{TP}{TP+FP}
$$

6.3 Recall

Recall measures the proportion of actual churn cases that were correctly identified.

$$
Recall=\frac{TP}{TP+FN}
$$

6.4 F1-Score

The F1-score is the harmonic mean of precision and recall.

$$
F1=2\frac{Precision\times Recall}{Precision+Recall}
$$

6.5 ROC-AUC

The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) measures how well the model distinguishes between the positive and negative classes across different probability thresholds.

An AUC closer to 1 indicates stronger class discrimination.

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7. RESULTS AND DISCUSSION

7.1 Final Model Performance

The final evaluation produced the following results:

Metric	Obtained Result

Test loss	0.2146
Test accuracy	91.39%
Test ROC-AUC	0.9669
Precision	69.14%
Recall	88.42%
F1-score	77.60%

The values above are from the model evaluation results obtained in Google Colab. Small differences may occur if the model is retrained with different settings or random states.

7.2 Interpretation of Results

Accuracy

The model achieved a test accuracy of 91.39%. This indicates that it correctly classified approximately 91.39% of the test examples at the selected classification threshold.

However, accuracy must be interpreted alongside the other metrics because the dataset contains more non-churn than churn customers.

Precision

The precision score was 69.14%. This indicates that approximately 69.14% of the customers predicted as churn cases were actually churn cases in the test set.

The remaining positive predictions were false positives. In a practical business setting, these predictions could result in retention actions being offered to customers who would not otherwise churn.

Recall

The recall score was 88.42%. This indicates that the model identified approximately 88.42% of the actual churn cases in the test set.

Recall is important when a business wants to identify as many potentially churning customers as possible.

F1-Score

The F1-score was 77.60%. It summarizes the balance between precision and recall. The result indicates that the model identifies many churn cases, although there is still room to improve the reliability of positive predictions.

ROC-AUC

The ROC-AUC score was 0.9669. This indicates strong discrimination between the churn and non-churn classes across classification thresholds on the test dataset.

ROC-AUC does not mean that 96.69% of individual predictions are correct. Accuracy measures the proportion of correct predictions at a selected threshold, while ROC-AUC measures discrimination across thresholds.

7.3 Overall Model Discussion

The model demonstrated promising performance for customer churn classification. Its recall was relatively high, suggesting that it identified a large proportion of the churn cases in the test dataset. Its lower precision indicates that some customers predicted to churn were actually in the non-churn class.

The appropriate balance between precision and recall depends on the intended business use. For example, a low-cost retention message may justify a greater emphasis on recall, whereas an expensive incentive campaign may require higher precision.

The results should be treated as an evaluation on the available test set, not as a guarantee of performance on future customers or on data collected from a different e-commerce platform.

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8. MODEL RESULT VISUALIZATIONS

The following visualizations were generated to examine the model's behavior and predictive performance.

8.1 Training Loss vs Validation Loss

Chart type: Line graph

Purpose: To compare the training and validation loss over the training epochs.

Interpretation: Training loss reflects how well the model fits the training data, while validation loss measures performance on unseen validation examples during training. A large and persistent gap between the two may indicate overfitting.

The final report should describe the actual trends visible in the generated chart, including whether validation loss stabilizes or increases.

8.2 Training Accuracy vs Validation Accuracy

Chart type: Line graph

Purpose: To examine changes in accuracy over training epochs.

Interpretation: Comparing the two curves helps determine whether the model improves on both training and validation data. A growing difference between training and validation accuracy can indicate that the model is fitting the training data more closely than the validation data.

The specific curve behavior should be described from the actual Colab output.

8.3 Confusion Matrix

Chart type: Heatmap

Purpose: To compare actual churn labels with predicted churn labels.

The confusion matrix contains four outcomes:

- **True Negative (TN):** A non-churn customer correctly predicted as non-churn.
- **False Positive (FP):** A non-churn customer incorrectly predicted as churn.
- **False Negative (FN):** A churn customer incorrectly predicted as non-churn.

- **True Positive (TP):** A churn customer correctly predicted as churn.

Interpretation: The confusion matrix reveals the types of classification errors made by the model. False negatives are particularly important when the objective is to identify customers who may leave.

The exact counts should be read from the generated confusion matrix.

8.4 ROC Curve

Chart type: Receiver Operating Characteristic curve

Purpose: To examine the relationship between the true positive rate and false positive rate at different classification thresholds.

Interpretation: A curve closer to the upper-left corner generally indicates stronger discrimination. The model's reported ROC-AUC was 0.9669, showing promising separation between the two classes in the test dataset.

8.5 Predicted Churn Probability Distribution

Chart type: Overlapping histograms

Purpose: To compare predicted churn probabilities for customers who actually stayed and customers who actually churned.

Interpretation: Less overlap between the two distributions suggests better separation. Overlap indicates that some customers from the two classes receive similar predicted probabilities and may be difficult to classify using the selected model.

8.6 Summary of Model Visualizations

Together, the training curves, confusion matrix, ROC curve, and predicted-probability distribution provide complementary information. Training curves help examine learning behavior, the confusion matrix summarizes errors at a chosen threshold, the ROC curve evaluates discrimination across thresholds, and the probability distribution shows how predictions differ between actual classes.

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9. CONCLUSION

This project demonstrated the application of exploratory data analysis, data preprocessing, visualization, and deep learning to e-commerce customer churn prediction.

The dataset contained 5,630 customer records and 20 columns describing customer tenure, device preferences, payment methods, order behavior, satisfaction, complaints, and churn status. The EDA process identified missing values in seven numerical features, confirmed that there were no exact duplicate rows, and revealed an imbalance between the churn and non-churn classes.

Several visualizations were used to examine numerical distributions, categorical features, correlations, potential outliers, and differences between churn groups. The preprocessing pipeline handled missing values, capped numerical outliers using training-set IQR limits, encoded categorical features, and standardized numerical inputs.

An MLP neural network was developed using TensorFlow/Keras. Basic hyperparameter tuning was performed using validation performance, and the selected model was evaluated on a separate test set.

The model achieved a test accuracy of 91.39%, precision of 69.14%, recall of 88.42%, F1-score of 77.60%, and ROC-AUC of 0.9669. These results suggest that the model can distinguish churn and non-churn customers effectively on the available test dataset, although the precision score indicates that some predicted churn cases are false positives.

Overall, the project illustrates how data analysis and deep learning can support customer-retention analysis. The model should be validated on future customer data before being used for real business decisions.

9.1 Future Scope

The project can be extended in the following ways:

1. **Advanced feature analysis:** Use permutation importance or SHAP to understand which customer attributes contribute most to predictions.
2. **Model comparison:** Compare the MLP against Logistic Regression, Random Forest, Gradient Boosting, and other tabular-data models.
3. **Threshold optimization:** Select a probability threshold that balances false positives and false negatives according to business requirements.
4. **Category standardization:** Merge equivalent labels such as CC and Credit Card before model training.
5. **Improved tuning:** Explore additional learning rates, hidden-layer sizes, dropout rates, and batch sizes.
6. **External validation:** Evaluate the model on newer customer data to test whether performance remains stable over time.
7. **Business integration:** Develop a dashboard that displays churn probabilities and supports the review of high-risk customer groups.
8. **Retention strategy evaluation:** Test whether targeted offers or customer-support interventions actually improve retention.

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10. REFERENCES

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APPENDIX A: IMPLEMENTATION DETAILS

The project was implemented in Google Colab using Python. The notebook contains the following code stages:

1. Importing the required libraries.
2. Loading the E Comm worksheet from the Excel dataset.
3. Inspecting dataset dimensions, columns, and data types.
4. Checking missing values and duplicates.
5. Cleaning column names and text values.
6. Plotting churn distribution and numerical feature histograms.
7. Visualizing categorical features and correlations.
8. Comparing numerical and categorical features with churn.
9. Detecting potential outliers.
10. Preparing the target and input features.
11. Splitting data into training, validation, and test subsets.
12. Building and applying the preprocessing pipeline.
13. Calculating class weights.
14. Creating and training the MLP neural network.
15. Comparing hyperparameter configurations.
16. Evaluating the final model.
17. Generating training curves, confusion matrix, ROC curve, and probability distribution.
18. Performing exploratory feature-weight analysis.
19. Saving the cleaned dataset, trained model, preprocessing pipeline, and evaluation metrics.

The complete executable code should be included in the submitted Google Colab notebook and linked from the GitHub repository.

APPENDIX B: EXPECTED PROJECT FILES

The GitHub repository should contain the following files or equivalent:

- README.md — project description, dataset source, setup instructions, and execution steps.
- Ecommerce_Customer_Churn_Analysis.ipynb — complete Google Colab/Jupyter notebook.
- report.pdf — final case study report.

- cleaned_ecommerce_data.csv — optional cleaned dataset, subject to dataset licensing and repository size limits.
- final_model_metrics.csv — final model evaluation metrics.
- hyperparameter_tuning_results.csv — validation results for the compared configurations.
- ecommerce_churn_model.keras — saved trained model, if appropriate for the repository.
- ecommerce_preprocessor.pkl — fitted preprocessing pipeline, if appropriate for the repository.
- visualizations/ — exported EDA and model evaluation charts.

The README should clearly state the source of the original dataset, the required Python libraries, and the steps needed to run the notebook. If the dataset cannot be redistributed, provide its source link instead of uploading the original file.